PL Insights: Data Cleaning & Transformation Report

Course: DL3004 Data Warehousing & Business Intelligence | Phase 2 Implementation | Date: November 2026

1. Executive Summary

This document details the data ingestion, cleansing, anomaly resolution, and transformation procedures implemented for the PL Insights Data Warehouse. The raw source comprises 16 Premier League seasons (2010/11 through 2025/26) obtained from football-data.co.uk (6,080 match records), coupled with a secondary dimensional lookup dataset containing stadium details and capacities. Cleansing and transformations were implemented strictly using reproducible PostgreSQL scripts in project_staging, ensuring a verifiable audit trail before promotion into the dimensional warehouse (project_dw).

2. Data Quality Issues Identified & Resolved

- **Inconsistent Date Formats:** Older seasons formatted dates as 2-digit years ('DD/MM/YY') while newer seasons used 4-digit years ('DD/MM/YYYY'). A regex-based conditional parser was applied in SQL to standardize all 2,280 two-digit records into valid ISO dates without data loss.
- **Entity Name Fragmentation:** Raw CSVs used abbreviated and colloquial team spellings (e.g., 'Man City', 'Wolves', 'Nott''m Forest', 'West Brom'). A comprehensive mapping table (team_name_map) standardized all 41 unique clubs to their official FA canonical names across 3,306 home match instances.
- **Match Grain Unpivoting:** Raw records store 1 row per match with home and away statistics side-by-side. To support granular team performance analysis, each match was unpivoted into exactly two team-centric observation rows, scaling the dataset from 6,080 matches to 12,160 team-match grain rows.
- **Missing / Blank Match Officials:** Historical matches occasionally lacked recorded referee names. Missing and blank values were systematically captured and coerced into 'Unknown' (mapping to surrogate key -1) to prevent orphan references.
- **Extraneous Betting Market Columns:** Raw feeds contained over 40 volatile betting market columns that varied across seasons. These were filtered out during initial ingestion, preserving solely operational match and disciplinary statistics.

3. Cleaning Audit Trail (staging.cleaning_log)

The transformation pipeline dynamically logged all cleansing metrics directly to PostgreSQL. The verified audit log is tabulated below:

Cleansing Metric / Action	Record Count	Validation Status
rows removed (corrupted/incomplete/duplicate)	**0**	VERIFIED PASS
raw rows ingested	**6080**	VERIFIED PASS
2-digit year dates converted	**2280**	VERIFIED PASS
NULL/empty referee mapped to Unknown	**0**	VERIFIED PASS
raw home team names remapped	**3306**	VERIFIED PASS
clean rows after transformation	**6080**	VERIFIED PASS
distinct standardized teams	**41**	VERIFIED PASS

4. Before vs. After Data Transformation Samples

Representative raw source rows contrasted against their transformed staging counterparts:

Field	Raw Staging Value (stg_matches)	Cleaned Staging Value (stg_matches_clean)	Transformation Logic
Date (Historical)	14/08/10	2010-08-14	Regex parse: to_date(..., 'DD/MM/YY')
Date (Recent)	16/08/2024	2024-08-16	Regex parse: to_date(..., 'DD/MM/YYYY')
Home Club Name	Man City	Manchester City	team_name_map lookup -> Canonical Name
Away Club Name	Wolves	Wolverhampton Wanderers	team_name_map lookup -> Canonical Name
Full Time Goals	'3' (text/varchar)	3 (integer)	NULLIF(TRIM(col), '')::integer
Disciplinary Cards	'2' / '' (text)	2 / 0 (integer)	Typecast with empty-string NULL coercion

5. Canonical Club Mapping Standard

41 Premier League clubs represented across the 16 seasons have been standardized and enriched with verified stadium infrastructure data (capacity, city, venue name) via team_lookup.csv.